Johannes Krotz

Curriculum Vitae - johanneskrotz.org

Applied and computational mathematics with interests in kinetic transport, uncertainty quantification, data assimilation, and scientific machine learning.

Education

2024 **PhD in Mathematics**, *University of Tennessee*, Knoxville, TN, Minor in Computational Sciences.
Dissertation: *Probabilistic and data-driven methods in numerical PDE*

2024 **Master of Science in Statistics**, *University of Tennessee*, Knoxville, TN.

2021 **Master of Science in Mathematics**, *Oregon State University*, Corvallis, OR.

2019 **Master of Science in Physics**, *University of Konstanz*, Konstanz, Germany, With honors.
Thesis: *Computer simulations of gel formation in colloidal systems of sticky rods*

2018 **Bachelor of Science in Mathematics**, *University of Konstanz*, Konstanz, Germany.
Thesis: *Gibt es rotationssymmetrische Fische: Existence proof for fish-shaped networks evolving under Gaussian curvature flow*

2015 **Bachelor of Science in Physics**, *University of Konstanz*, Konstanz, Germany.
Thesis: *Quantum transport in topological insulator-superconductor heterostructures*; conducted at **Reykjavik University, Reykjavik, Iceland**

2012 **Abitur (higher education entrance qualification)**, *Friedrich-Woehler Gymnasium*, Singen, Germany.

Honors, Awards and Grants

2025 Elected Full Member, Sigma Xi, The Scientific Research Honor Society

2024 Third place UTK SIAM research showcase

2024 Inducted Member, The Honor Society of Phi Kappa Phi

2023 First place UTK SIAM research showcase

2023 Academic Excellence Fellowship (UTK Math)

2023 UTK Graduate Student Senate Travel Awards

2022 Randall E. Cline Awards

2022 National Science Foundation Mathematical Sciences Graduate Internship (NSF MSGI)

2022 Academic Excellence Fellowship (UTK Math)

2021 F. M. Dryzer Memorial Graduate Fellowship

2021 National Science Foundation Mathematical Sciences Graduate Internship (NSF MSGI)

2020 National Science Foundation Mathematical Sciences Graduate Internship (NSF MSGI)

2018 Second place in LöPhi awards for excellent teaching (awarded through the physics student union according to student vote)

Research and Professional Experience

2024–present Postdoctoral Researcher, Department of Aerospace and Mechanical Engineering, University of Notre Dame. Developing hybrid Monte Carlo/deterministic kinetic transport algorithms in collaboration with the Center for Advancing the Radiation Resilience of Electronics and the Center for Exascale Monte Carlo Neutron Transport.

2023 **GRA funded through ORNL**. Dynamic-likelihood filtering for stochastic advection-diffusion equations with Juan Restrepo and Jorge Ramirez.

2022 **Research Intern at ORNL – Mathematics in Computation funded through NSF MSGI**. Hybrid algorithms for simulating complex particle systems in 2D & 3D with Cory Hauck and Ryan McClarren.

2021 **Research Intern at ORNL – Mathematics in Computation funded through NSF MSGI**. Hybrid algorithms for simulating complex particle systems in 1D with Cory Hauck and Ryan McClarren.

2020 **Research Intern at LANL – Computational Earth Science funded through NSF MSGI**. Fast 2D and 3D Poisson-disk sampling algorithms for high-quality triangulations of discrete fracture networks with Carl Gable, Jeffrey Hyman, and Matthew Sweeney.

2019 **Master's student in AG Nielaba – Statistical and Computational Physics (Konstanz)**. Computer simulations of colloidal systems and anisotropic particles, including clustering, TDA, and shear-stress analysis.

2018 **Bachelor's student in AG Schnuerer – Differential Geometry**. Work on the existence of solutions to Gaussian and mean curvature flows under symmetry constraints.

2015 **Exchange scholar at Reykjavik University & Bachelor's student in AG Belzig – Quantum Transport**. Theoretical work on topological insulator-superconductor systems with Erlingsson Sigurður Ingi.

Publications

1 *Generative Monte Carlo Sampling for Constant-Cost Particle Transport*, Joseph Farmer, Aidan Murray, **Johannes Krotz**, Ryan G. McClarren, under review at *PNAS*.

2 *Scatter-Limited Hybrid Monte Carlo, Deterministic Transport with Quasi–Monte Carlo Sampling*, **Johannes Krotz**, Ryan G. McClarren, to appear in *Proceedings of PHYSOR 2026*.

3 *Curvilinear coordinates and curvature in radiative transport*, **Johannes Krotz**, Ryan G. McClarren, arXiv:2508.20852; submitted Aug. 28, 2025; under review at *Journal of Computational and Theoretical Transport*.

4 *A dynamic likelihood approach to filtering transport processes: advection-diffusion dynamics*, **Johannes Krotz**, Juan M. Restrepo, Jorge Ramirez, *Journal of Computational Physics*, Volume 536 (2025).

5 *A Hybrid Monte Carlo, Discontinuous Galerkin method for linear kinetic transport equations*, **Johannes Krotz**, Cory D. Hauck, Ryan G. McClarren, *Journal of Computational Physics*, Volume 514 (2024).

6 *Variable resolution Poisson-disk sampling for meshing discrete fracture networks*, **Johannes Krotz**, Matthew R. Sweeney, Carl W. Gable, Jeffrey D. Hyman, Juan M. Restrepo, *Journal of Computational and Applied Mathematics*, Volume 407 (2022).

Presentations

Highlighted talks

2026 *Scatter-Limited Hybrid Monte Carlo, Deterministic Transport with Quasi–Monte Carlo Sampling*, accepted for presentation at PHYSOR 2026, Turin, Italy

2026 *Generative Models for Physical Processes*, presented at the 2026 MICDE Predictive Science Symposium, Ann Arbor, MI

2025 *Normalizing Flow-Based Monte Carlo Transport Simulation*, presented at the International Conference on Monte Carlo Methods and Applications (MCM 2025), Chicago, IL, USA

2025 *Hybrid Monte Carlo Methods for Kinetic Transport*, presented at the International Conference on Monte Carlo Methods and Applications (MCM 2025), Chicago, IL, USA

2025 *Minimizing Uncertainty in Transport Problems through Dynamic Likelihood Filtering*, presented at the Southeastern Atlantic Section of the SIAM Section Meeting 2025, Knoxville, TN

2024 *Hybrid Method for Kinetic Transport Equations*, presented at the SIAM New York–New Jersey–Pennsylvania Section Conference, Rochester, NY, USA

2023 *A Hybrid Monte Carlo, Discontinuous Galerkin method for linear kinetic transport equations*, presented at the 6th Annual SIAM Texas–Louisiana Sectional Meeting, Lafayette, LA

Other talks and posters (posters marked with dagger)

2025 *Computational Hybrid Methods for Kinetic Transport*, presented at the COSE Research Horizons Symposium, University of Notre Dame

2025 *MCDG for MCDC - Hybrid Monte Carlo Discontinuous Galerkin Methods for Exascale Transport*, presented at the 2025 PSAAP III Final Review Meeting, Raleigh, NC

2025 *Minimizing Uncertainty in Transport Problems through Dynamic Likelihood Filtering*, presented at the 2025 AMS Spring Central Sectional Meeting, Lawrence, KS

2024 *A Hybrid Monte Carlo, Discontinuous Galerkin method for linear kinetic transport equations*, presented at the Center for Exascale Monte Carlo Neutron Transport

2024 *A Likelihood Approach to Filtering for Advection Diffusion Processes*, presented at the 103rd Annual MAA Southeastern Section Meeting, Knoxville, TN

2025 *Introduction to Bayesian Filtering and Applications in Transport Problems*, presented at the Applied Math and Computation Seminar, Oregon State University

2024 *Efficient solvers for kinetic transport equations*, presented at the University of Konstanz, Colloquium on Numerical Optimization

2023 *Efficient solvers for the neutron transport equation*, presented at Karlsruher Institut für Technologie (KIT), Oberseminar (Comp. Science and Mathematical Methods)

2025 †*Dynamic Likelihood Filtering - Data Assimilation for Advective Transport*, presented at Data Assimilation and Inverse Problems for Digital Twins (IMSI/Illinois Institute of Technology)

2025 †*Conditional Normalizing Flows for Transport Simulation*, presented at Statistical and Computational Challenges in Probabilistic Scientific Machine Learning (SciML), Chicago

2025 †*Minimizing Uncertainty in Transport Problems through Dynamic Likelihood Filtering*, presented at Uncertainty Quantification and Machine Learning for Complex Physical Systems (IMSI/Illinois Institute of Technology)

2025 † *A Dynamic Likelihood Approach to Filtering*, presented at the 2025 PSAAP III Final Review Meeting, Raleigh, NC

2024 † *A Dynamic Likelihood Approach to Filtering for Advection-Diffusion Dynamics*, presented at Robust Optimization and Simulation of Complex Stochastic Systems (ICERM), Providence, RI

2023 *A Hybrid Monte Carlo, Discontinuous Galerkin method for linear kinetic transport equations*, presented at Summer School on Computational & Data Science at Duke University

2023 † *A Hybrid Monte Carlo, Discontinuous Galerkin method for linear kinetic transport equations*, presented at the University of Chicago AI+Science Summer School, Chicago, IL

2022 † *Hybrid methods for the numerical simulation of complex particle systems*, presented at 2022 NSF MSGI Summer Research Symposium

2020 *Dataset reduction in TDA with an example from soft matter physics*, presented at the Data Science and Image Analysis Conference of the Pacific Northwest, Pullman, WA

2019 †*Gel formation in colloidal systems of sticky rods*, SFB 1214: Anisotropic Particles as Building Blocks, Konstanz, Germany

2018 †*Computer simulation of anisotropic colloidal particles*, Symposium on Anisotropic Particles, Konstanz, Germany

Service

Conference and Professional Service

2026 Organizer of a minisymposium on *Stochastic and Data-Driven Computation in Science and Engineering* at the SIAM Annual Meeting 2026

2026 Panelist for the "Session on AI and Machine Learning" at the 2026 MICDE Predictive Science Symposium

2025 Organizer of a minisymposium on *Applications and Algorithms for Diff. Eq. in Modelling and Engineering* at the Southeastern Atlantic Section of the SIAM Section Meeting 2025

Editorial and Reviewing

2026 Reviewer for Nonlinear Processes in Geosciences

2025 Guest Editor for a special issue of Journal of Computational and Theoretical Transport

2025 Reviewer for several papers in Journal of Computational and Theoretical Transport

2025 Reviewer for Journal of Sound and Vibration

2025 CRC Press (Taylor & Francis Group) — Book Proposal Reviewer

Academic and Community Service

2026 Judge at Northern Indiana Regional Science and Engineering Fair

2025 Judge at COSE Research Horizons Symposium

2025 Judge, Northern Indiana Regional Science and Engineering Fair

2023-2024 Elected member of UTK's Math Grad Student Council (MGSCI)

2021-2022 Elected member of UTK's Graduate Student Senate

Community Leadership

2010-2020 Elected board member Musikverein Bohlingen (Orchestra)

- 2018-2020 **2nd Chairman**: Coordination and organization of concerts, fundraisers, festivals grossing up to $\approx 300000\$$,
- 2013-2018 **Youth coordinator**: Coordinating youth orchestra, finding/hiring instrumental teachers and conductors, organizing merger and inclusion of 2 neighboring youth orchestras, organizing concerts and retreats

Professional Memberships

- Society for Industrial and Applied Mathematics (SIAM)
- American Mathematical Society (AMS)
- Association of Women in Mathematics (AWM)
- American Association for the Advancement of Science (AAAS)
- Society for the Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS)

Programming & Computational Skills

Scientific Computing:

- **Python** – Primary language for research. Developed simulation pipelines, hybrid Monte Carlo solvers, stochastic interpolants, and DNN models. Contributed to `MCDC`, a CUDA-compatible exascale neutron transport code using `Numba`. Created a mesh generation module now used in `DFNWorks` for porous media flow and PDEs.
- **C/C++** – Maintained and extended a large colloidal particle simulation package as part of my M.S. thesis. Integrated OpenMP-based parallelism for HPC execution with moderate success. Frequently interfaced with Fortran in large-scale simulation frameworks.
- **Matlab** – Used for 2D radiation transport solvers combining MC and discrete ordinates. Implemented PDE solvers, uncertainty quantification algorithms (e.g., Kalman-type filters), and completed coursework in Bayesian inference, optimization, and numerical analysis.
- **Fortran** – Read and interfaced with legacy Fortran code in LANL's `DFNWorks`. Used for small solvers and coursework.

Tooling and Environment:

- **High-Performance Computing**: Frequent use of HPC systems for running large-scale particle and PDE simulations.
- **Version Control & Shell**: Proficient in `Git`, `Bash`, and typical HPC workflows
- **LaTeX**: Over 10 years of experience, including Beamer and TikZ
- **Visualization**: Experienced with `Matplotlib`, `Gnuplot`, `Matlab`, and Python-based scientific plotting.

Languages

German	**Native**	
English	**Fluent**	*6+ years of academic work in the US; 115/120 on TOEFL in 2019*
Italian	**Conversational**	*5 years of instruction; functional with immersion*

Teaching

Instructor of record where indicated; otherwise teaching assistant.

University of Notre Dame

2024–present	Co-advisor in weekly meetings, guiding the research of Ph.D. students and undergraduate students.

University of Tennessee Knoxville

Spring 2023	Instructor of Record for Finite Math (MTH 123)
Fall 2022	Instructor of Record for Finite Math (MTH 123)
Spring 2022	Instructor of Record for Basic Calculus (MTH 125)
Fall 2021	Basic Calculus (MTH 125)

Oregon State University

Spring 2021	Advanced Calculus II (MTH 312)
Winter 2021	Advanced Calculus I (MTH 311)

Fall 2020	Linear Algebra II (MTH 342)
Spring 2020	Linear Algebra I (MTH 341)
Winter 2020	College Algebra (MTH 111)
Fall 2019	Elementary Functions (MTH 112)
	University of Konstanz
Summer 2019	Real Analysis II (Department of Mathematics)
Summer 2019	Linear Algebra and Analysis for Computer Scientists (Department of Computer and Information Science)
Winter 18/19	Theoretical Physics III (Department of Physics)
Winter 18/19	Mathematics for Physicists III (Department of Mathematics)
Summer 2018	Discrete Mathematics (Department of Computer and Information Science)
Summer 2018	Mathematics for Physicists II (Department of Mathematics)
Winter 17/18	Mathematics for Physicists III (Department of Mathematics)
Summer 2017	Mathematics for Physicists II (Department of Mathematics)
Winter 16/17	Mathematics for Physicists I (Department of Mathematics)
Summer 2016	Discrete Mathematics (Department of Computer and Information Science)